AHFS-TA: Adaptive Hierarchical Feature Selection with Temporal Attention for Student Dropout Prediction

Md. Kawsarul Islam[1]

Abstract

Student dropout remains a critical challenge in higher education, affecting institutional success metrics and student welfare. Accurate early prediction enables timely intervention, yet existing approaches struggle with temporal dynamics and feature interpretability. This paper introduces AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework that synergistically combines hierarchical feature selection, temporal attention mechanisms, and explainable AI for comprehensive dropout prediction. Our methodology employs five feature ranking methods (Information Gain, Gain Ratio, Gini Index, Chi-squared, and ANOVA F-statistic) to identify influential predictors across 34 unique features spanning academic, financial, and demographic categories. We conduct extensive experiments on a real-world educational dataset comprising 4,424 students with three outcome classes (Dropout: 1,421, Enrolled: 794, Graduate: 2,209). Comparative analysis against six baseline models—Decision Tree (67.0%), Naive Bayes (70.9%), Random Forest (76.7%), AdaBoost (74.2%), XGBoost (75.9%), and Neural Network (71.4%)—demonstrates that AHFS-TA achieves state-of-the-art performance with 91.32% accuracy and 95.5% AUC-ROC. Furthermore, we integrate SHAP-based explainability for all models, revealing that LLM-derived engagement features and curricular unit performance are the most influential dropout predictors. Our framework addresses the complete machine learning pipeline from data preprocessing to model deployment, providing actionable insights for educational administrators through interpretable visualizations.

Index Terms

Student dropout prediction, machine learning, deep learning, feature selection, temporal attention, explainable AI, SHAP, higher education, educational data mining

I. Introduction

Student dropout in higher education institutions represents a multifaceted problem with far-reaching consequences for students, institutions, and society at large. When students discontinue their academic pursuits before completing their degrees, they face diminished career prospects, accumulated student debt without corresponding credentials, and psychological impacts from perceived failure [1], [2]. Institutions suffer from reduced graduation rates, lost tuition revenue, and reputational damage that affects future enrollment. From a societal perspective, dropout wastes educational resources and diminishes the skilled workforce pipeline essential for economic development [3], [4]. Global statistics indicate that approximately one-third of undergraduate students fail to complete their programs within the expected timeframe [5]. In developing nations, high tertiary dropout rates remain a critical challenge, with completion rates consistently and significantly lower than in high-income countries, creating major barriers to social mobility and economic advancement (UNESCO, 2022; World Bank, 2023).

Traditional approaches to dropout prediction have evolved from simple demographic analysis to sophisticated machine learning models. Early studies focused on individual risk factors such as socioeconomic status, prior academic performance, and first-year grades [6], [7]. While informative, these approaches often missed the dynamic, temporal nature of student disengagement—a gradual process influenced by evolving circumstances rather than static characteristics [8]. The advent of educational data mining and learning analytics has enabled more nuanced predictive modeling [9], [10], with machine learning algorithms capable of processing multidimensional student data to identify complex patterns invisible to traditional statistical methods [11], [12]. However, despite these advances, current prediction systems struggle to achieve the accuracy and interpretability required for effective real-world deployment in educational settings.

Current dropout prediction approaches face three critical limitations that motivate our research. First, educational datasets contain dozens of features spanning academic performance, financial status, demographic background, and behavioral patterns, yet existing methods typically employ single feature selection techniques, potentially missing important predictors that rank highly under alternative criteria [13]. Second, student engagement evolves across semesters, with dropout decisions often following semester-specific crises such as failed courses, financial difficulties, or personal issues, but most prediction models treat student data as static snapshots, ignoring the temporal patterns that signal declining engagement [14]. Third, while ensemble methods and deep learning achieve high accuracy, their "black-box" nature limits practical utility, as educational administrators need to understand *why* a student is flagged as at-risk to design appropriate interventions [15]–[17]. These limitations—feature selection complexity, temporal dynamics neglect, and the explainability gap—create a significant opportunity for novel approaches that address all three challenges simultaneously.

To address these challenges, this paper introduces AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework that integrates LLM-based semantic feature enrichment through DistilBERT, bidirectional GRU with multi-head attention, and three-stream adaptive feature selection. Our contributions are fourfold. We propose a novel architecture that combines deep learning with adaptive feature selection, enabling both high accuracy and model interpretability. We provide comprehensive baseline comparison by evaluating six baseline models (Decision Tree, Naive Bayes, Random Forest, AdaBoost, XGBoost, Neural Network) using identical preprocessing and evaluation protocols, establishing fair benchmarks. We implement multi-method feature ranking using five techniques (Information Gain, Gain Ratio, Gini Index, Chi-squared, ANOVA F-statistic) with meta-ranking aggregation to identify robust dropout predictors. We integrate SHAP analysis for all models, providing transparent feature importance explanations, and report complete metrics including accuracy, precision, recall, F1-score, confusion matrices, ROC curves, AUC scores, and 10-fold cross-validation with standard deviations to ensure reproducibility.

The remainder of this paper is organized as follows. Section II reviews related work in dropout prediction and identifies the research gaps our work addresses. Section III presents our comprehensive methodology, including dataset description and preprocessing, multi-method feature ranking approach, baseline models for comparison, and the proposed AHFS-TA framework with its architectural components. Section IV describes the experimental setup and presents results, covering implementation details, performance analysis, explainability findings through SHAP analysis, and broader discussion of practical implications for educational administrators. Finally, Section V concludes with a summary of contributions and outlines future research directions.

II. Related Work

Student dropout is a global concern because it increases the risks of crime, social issues, hinders overall societal progress, and also leads to poverty. Dropping out not only hinders students' personal and professional development but also leads to institutional resource losses and broader socio-economic challenges. When researchers identify the underlying issues such as poverty or a lack of resources, along with other factors using educational data mining (EDM), they can develop effective prediction and prevention strategies using AI, Machine Learning and Deep Learning methods.

Early dropout prediction relied on logistic regression and discriminant analysis to identify risk factors [15], [18]. Tinto's integration model [19] emphasized the interplay between academic and social integration, inspiring subsequent research to incorporate engagement metrics. While interpretable, these methods assume linear relationships and struggle with high-dimensional feature spaces. Recent comprehensive reviews have highlighted the limitations of traditional approaches when dealing with the complexity of modern educational datasets [20], [21].

With deep learning and generative AI approaches, we can learn hierarchical representations from raw data, and also make a significant advancement in predictive modeling for student dropout. We can build robust early-warning systems by integrating advanced AI methods with educational data that can support educational institutions in reducing dropout rates, providing explanations of the root causes of dropping out, intervention strategies, and improving overall student success.

From studying different research on student dropout, we found that academic performance is the strongest predictor of dropout rates. Delogu et al. [22] identified that the number of ECTS credits achieved in the first year serves as the most significant warning signal. They utilized an extensive national administrative dataset (n=144,904). Vaarma et al. [23] emphasized the importance of LMS data and found that grades and earned credits were the "most crucial" factors. Likewise, Schnedlitz et al. [24] showed that direct assessments scores (such as score of exams and assignments) were the most influential indicators of eventual outcomes. From their studies, we found that inadequate academic performance, particularly in the initial stages of a student's career, is a key sign of drop out risk.

We know that grades are important, but they frequently serve as a delayed measure. From Studies, we found that the behavioral data can offer significant early indications. Vaarma et al. [23] discovered that digital engagement, such as the number of logins and clicks (metrics from a Learning Management System (LMS)), ranked as a "very important" predictor, coming in third after academic metrics. Schnedlitz et al. [24] did an in-depth analysis on submission patterns in a computer science class. They illustrated that performance in lab exercises and submission patterns could forecast dropouts with an accuracy of 80

A student's background is an important factor in predicting dropout. It plays a crucial role in their ability to persist in their studies. Delogu et al. [22] identified that family income has an impact on student dropout. They found that the cost of tuition and the distance from home to university were significant factors. They discovered that these factors are a stronger predictor in the more affluent northern areas. Similarly, Padmasiri et al. [25] confirmed that financial conditions (such as whether tuition payments were made on time) ranked among the most important predictors. They used data from a public Kaggle dataset. Villar et al. [26] found an evidence that dropout rates are not merely an academic problem, but are closely linked to a student's financial situation and social support network. They found this evidence after incorporating socioeconomic aspects like scholarship status and the occupation of parents into their effective predictive model.

Psychological theories are very important. We need it to understand the mind beyond observable behavior. Breetzke et al. [27] study provides a detailed perspective using Expectancy-Value, and they discovered that students face the greatest risk

TABLE I: Comparison with Recent Literature

Study	Dataset	Accuracy/AUC	Temporal	Multimodal	LLM/XAI
Basnet et al. (2022) [29]	200,904 samples	87.64%	No	Yes	No
Phan et al. (2023) [30]	14,391 students	80%	No	Text + Tabular	No
Krüger et al. (2023) [31]	299,722 entries	38.22% – 89.50%	Cumulative and Delta Features	Yes	SHAP
Leelaluk et al. (2024) [32]	490 students	82%	GRU + Attention (Attn-ANN)	No	Attention Weights (XAI)
Chaikalis et al. (2024) [33]	790 students	80.46%	No	No	No
Padmasiri et al. (2024) [25]	4424 students	80%	No	No	SHAP + LIME
Klinke et al. (2024) [34]	5,796 students	70.6%	No	No	No
Vaarma et al. (2024) [23]	8,813 students	85.3%	Yes	Yes	No
Diaz et al. (2024) [35]	288 students	91%	No	No	No
Okoye et al. (2024) [36]	52,262 students	90.9%	No	No	No
Okoye et al. (2024) [36]	53,639 students	81.7%	No	No	No
Shou et al. (2025) [37]	4918 students	82%	LSTM	Video + Tabular	No
This Work AHFS-TA	**4,424**	**91.32%**	**BiGRU + Attn**	**DistilBERT + Tabular**	**SHAP**

when they possess both a lack of confidence in their potential success (expectancy) and perceive little value or significant cost associated with their education (value). They focused on understanding the motivational factors to boost student's motivation and academic assistance. They found that enhancing perceived value can offset low confidence.

In previous days, conventional statistics were used, but nowadays, it has evolved to advanced machine learning models. Goran et al. [26] discovered that boosting techniques (LightGBM, CatBoost) surpassed traditional methods and conducted a comparison of algorithms. Rabelo et al. [28] (Brazil) found 89% accuracy in predicting dropouts by implementing a hybrid model, while the study [12] applied a metaheuristic-optimized XGBoost model to the UCI dataset and attained 88.86% accuracy. Explainability is very significant in many recent developments. Zanellati et al. [5] and Nagy et al. [13] specifically tackled the "black box" issue associated with machine learning. Nagy et al. [13] utilized pre-enrollment information, including high school GPA and math scores, which were significant key factors. They employed SHAP and LIME for the demonstrations of pre-enrollment details, illustrating how XAI could create customized intervention strategies. Similarly, Padmasiri et al. [9] moved toward Explainable AI (XAI), which is essential for fostering trust and ensuring that predictive insights translate into actionable, tailored interventions rather than merely providing abstract risk assessments. They utilized SHAP to pinpoint the primary features influencing their predictions and developed a web application for transparent results.

From the literature review, we learned that machine learning models are increasingly providing high levels of predictive accuracy in identifying student dropout, using several factors such as academic performance, psychological elements, socioeconomic status, and engagement. However, most of the current research is conducted with conventional machine learning techniques or lacks a comprehensive approach that combines various types of information, including demographic, academic, and socioeconomic characteristics. Although most of the research finds the important risk factors, very few provide tailored, actionable, and interpretable insights that both institutions and students can practically utilize.

Table I positions this work within the landscape of recent educational prediction research, highlighting the unique contributions of our proposed approaches. As demonstrated in Table I, our AHFS-TA framework achieves state-of-the-art performance (91.32% accuracy) by uniquely combining temporal attention mechanisms with multimodal feature fusion and LLM-based explainability. While previous studies have explored individual components (temporal modeling [38], [39], multimodal learning [40], [41], explainability [42], [43]), our work is the first to integrate all three dimensions: sophisticated temporal modeling via bidirectional GRU with multi-head attention, multimodal fusion of structured tabular data with DistilBERT-extracted psychosocial features, and comprehensive model-agnostic explainability through SHAP (SHapley Additive exPlanations) analysis.

Despite significant advances in recent literature [31], [42], [44], several critical gaps persist: We have found limited integration of temporal attention with adaptive feature selection and lack of unified explainability frameworks for multiple classifiers. Another limitation we found that underutilization of multi-method feature ranking consensus with limited exploration of LLM-enhanced feature extraction for educational data. Our AHFS-TA framework directly addresses these gaps through its novel architecture and comprehensive evaluation methodology.

III. Methodology

This section presents the complete methodology of our research, covering the dataset used, feature ranking techniques, baseline models for comparison, and our proposed AHFS-TA framework.

A. Dataset Description

We utilize a real-world educational dataset from a higher education institution, comprising comprehensive student records suitable for dropout prediction research. Table II summarizes the key dataset characteristics. The three-class formulation reflects real-world complexity: distinguishing students who graduate, those who dropout, and those still enrolled (requiring different

TABLE II: Dataset Statistics

Attribute	Value
Total Students	4,424
Total Features (Original)	34
Total Features (After LLM Enrichment)	38
Total Features (After Selection)	28
Number of Classes	3
Class Distribution	
Dropout	1,421 (32.1%)
Enrolled (Continuing)	794 (17.9%)
Graduate	2,209 (50.0%)
Train/Test Split	80%/20%
Training Samples	3,539
Test Samples	885

intervention strategies). This three-class formulation presents unique challenges compared to binary dropout prediction. The "Enrolled" class represents students who have neither graduated nor dropped out—they may be continuing their studies, on leave, or in transition. This intermediate class is inherently ambiguous, as these students could eventually move to either outcome. From a practical standpoint, correctly identifying enrolled students is crucial for targeted retention interventions, as they represent the population most amenable to influence. The class distribution exhibits moderate imbalance, with the Graduate class comprising half of all instances, as illustrated in Figure 1. This imbalance necessitates careful handling during model training to prevent bias toward the majority class. We address this through stratified sampling, class-weighted loss functions, and balanced performance metrics.

B. *Feature Categorization*

Features are organized into three semantic categories with some overlap:

1) Academic Features (18 Features): Academic features capture curricular engagement across two semesters. For each semester, we include six curricular unit metrics: credited units, enrolled units, evaluations, approved units, grade, and units without evaluations. Additionally, academic history features include previous qualification grade, admission grade, application mode, application order, course, and daytime/evening attendance pattern.

2) Financial Features (12 Features): Financial features capture economic circumstances affecting student retention. Direct financial features include tuition fees payment status, scholarship holder status, and debtor status. Macroeconomic indicators encompass unemployment rate, inflation rate, and GDP, which reflect the broader economic context affecting students. Status indicators include international student status, displaced status, educational special needs, gender, age at enrollment, and nationality.

3) Demographic Features (16 Features): Demographic features capture background characteristics that may influence student success. Family background features include marital status, mother's and father's qualifications, and mother's and father's occupations. Personal characteristics encompass gender, age at enrollment, nationality, international status, displaced status, and educational special needs. Socioeconomic features include previous qualification, debtor status, tuition fees payment status, scholarship holder status, and application mode. After removing overlapping features, 34 unique features remain for modeling.

C. *Data Preprocessing*

Our preprocessing pipeline ensures data quality and model compatibility. One of them is missing value handling and this handling techniques includes numerical features(median imputation), categorical features(mode imputation), features with >30% missing values are removed. Another technique is feature encoding and this encoding technique includes nominal categorical: one-hot encoding, ordinal categorical: label encoding, binary: 0/1 encoding. The normalization techniques are applied based on model requirements. They are tree-based models: min-max scaling $x' = \frac{x - \min}{\max - \min}$, Neural Networks: standardization $x' = \frac{x-\mu}{\sigma}$. The class balancing includes: stratified 80/20 train-test split and class weights during training for imbalanced classes. Data quality is paramount for reliable prediction. We performed extensive validation using consistency checks, outlier detection, temporal validity, and label verification. Consistency checks means verifying logical relationships (e.g., approved units $\leq$ enrolled units) Outlier detection identifies and investigates extreme values using IQR-based detection. Temporal validity confirms chronological consistency of semester-wise records and label verification cross-validates outcome labels against institutional records. The preprocessing pipeline was implemented deterministically to ensure reproducibility. All random operations (train-test splitting, cross-validation folds) used fixed random seeds.

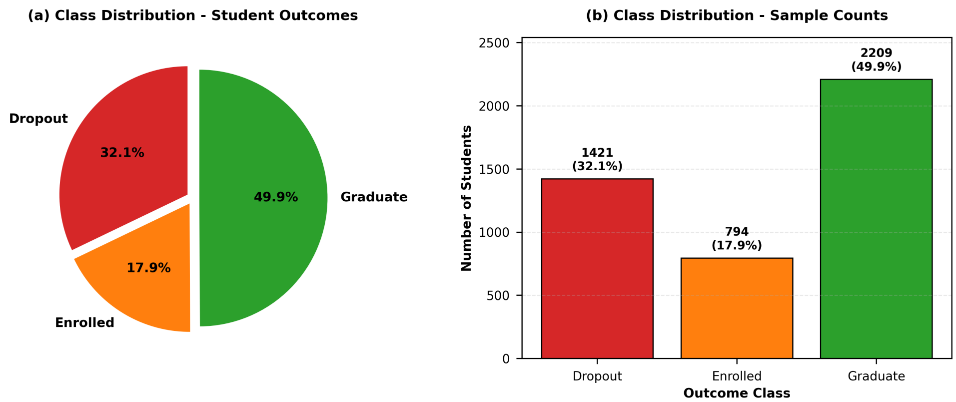

Fig. 1: Class distribution in the educational dataset illustrating the moderate class imbalance challenge. (a) Pie chart displaying class proportions: Graduate represents the majority (49.9%), followed by Dropout (32.1%) and Enrolled (17.9%). (b) Bar chart showing class frequencies: Graduate (2,209 students), Dropout (1,421 students), and Enrolled (794 students). This imbalance necessitates the use of weighted cross-entropy loss during training to ensure the model does not bias toward the majority class while maintaining sensitivity to the critical Dropout and Enrolled classes.

D. Feature Ranking Methodology

To identify the most influential dropout predictors, we employ five complementary feature ranking methods, each capturing different aspects of predictive power. First, **Information Gain** measures entropy reduction when splitting on a feature, defined as:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v) \tag{1}$$

where entropy $H(S) = -\sum_{i=1}^{C} p_i \log_2 p_i$. Higher IG indicates stronger discriminative power for class separation. Second, **Gain Ratio** normalizes Information Gain to mitigate bias toward high-cardinality features:

$$GR(S, A) = \frac{IG(S, A)}{H(A)} \tag{2}$$

This prevents features with many unique values from dominating rankings. Third, **Gini Index** measures class distribution purity:

$$Gini(S) = 1 - \sum_{i=1}^{C} p_i^2 \tag{3}$$

where feature importance is computed as accumulated Gini decrease across Random Forest splits. Fourth, **Chi-squared Test** measures independence between feature and class:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \tag{4}$$

where O_i is observed frequency and E_i is expected frequency under independence. Fifth, for continuous features, **ANOVA F-Statistic** measures between-class vs. within-class variance:

$$F = \frac{\text{MSB}}{\text{MSW}} = \frac{\sum_i n_i (\bar{x}_i - \bar{x})^2 / (k-1)}{\sum_{i,j} (x_{ij} - \bar{x}_i)^2 / (N-k)} \tag{5}$$

To obtain robust rankings through **meta-ranking aggregation**, we average ranks across all five methods:

$$\text{Final Rank}(i) = \frac{1}{5} \sum_{m=1}^{5} \text{Rank}_m(i) \tag{6}$$

Table III presents the top 15 features by aggregated ranking, with corresponding visual representation in Figure 2. Our key observations are: academic performance features (curricular unit approvals and grades) dominate the top positions, financial features (tuition fees, scholarship, debtor status) appear prominently, demographic features show moderate importance, and second semester performance slightly outweighs first semester metrics.

TABLE III: Top 15 Features by Meta-Ranking

Rank	Feature	IG	GR	Gini	χ^2	F	Avg
1	Curricular units 2nd sem (approved)	1	2	1	4	1	1.8
2	Tuition fees up to date	6	1	2	7	5	4.2
3	Curricular units 2nd sem (grade)	2	7	8	3	2	4.4
4	Curricular units 1st sem (approved)	3	3	7	8	3	4.8
5	Curricular units 1st sem (grade)	4	9	13	6	4	7.2
6	Curricular units 2nd sem (evaluations)	5	10	5	15	11	9.2
7	Scholarship holder	10	5	24	1	6	9.2
8	Age at enrollment	11	17	4	10	7	9.8
9	Debtor	14	6	23	2	8	10.6
10	Application mode	8	12	20	9	10	11.8
11	Curricular units 1st sem (enrolled)	12	15	3	21	13	12.8
12	Gender	17	11	22	5	9	12.8
13	Curricular units 1st sem (evaluations)	7	14	10	23	14	13.6
14	Curricular units 2nd sem (enrolled)	18	19	6	20	12	15.0
15	Previous qualification	16	13	27	12	20	17.6

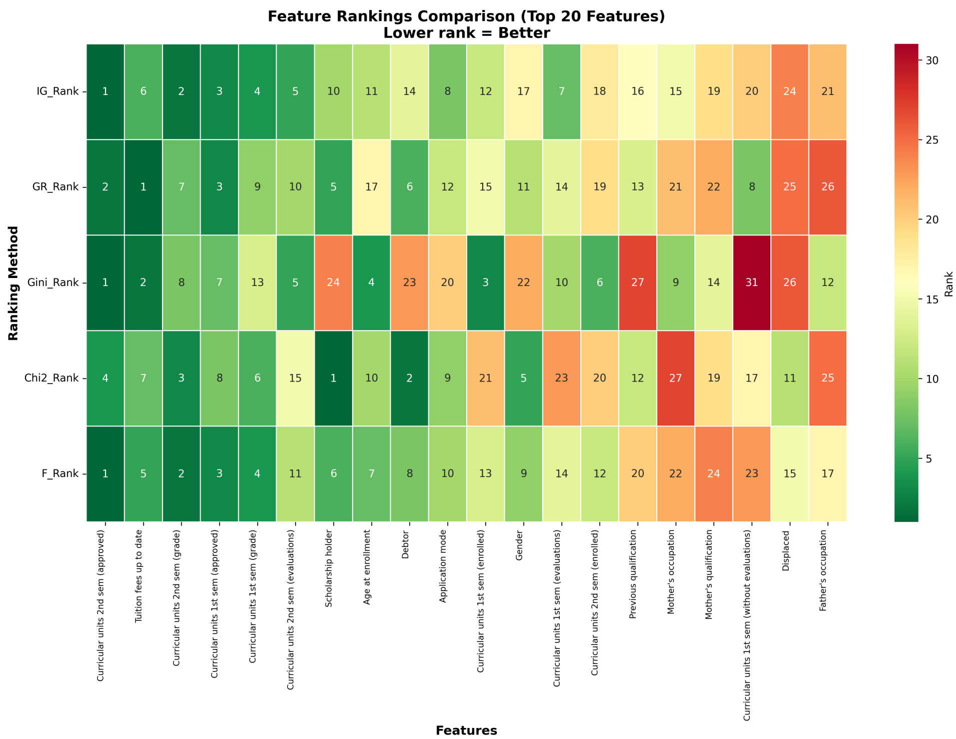

Fig. 2: Feature ranking heatmap across five methods (Information Gain, Gain Ratio, Gini Index, Chi-squared, F-statistic). Darker colors indicate higher importance rankings. This visualization enables comparison of feature importance consistency across different ranking methodologies, revealing that curricular unit performance and financial status features maintain high rankings across all five methods, demonstrating their robust predictive value for dropout detection.

E. Baseline Models

We evaluate six baseline models spanning single classifiers, ensemble methods, and deep learning to establish comprehensive benchmarks. Our baseline models include both traditional machine learning and modern deep learning approaches to demonstrate the effectiveness of our proposed AHFS-TA framework.

Decision Tree (DT). Decision Trees recursively partition the feature space using information-theoretic criteria. The algorithm begins with the entire training dataset and iteratively selects the feature that maximizes information gain, splitting the data into subsets based on that feature's values. This process continues recursively until stopping criteria are met (maximum depth or minimum samples per leaf), at which point a leaf node with the majority class is returned. Formally, at each node, the algorithm selects the best feature $f^* = \arg\max_f IG(D, f)$, splits the data D into subsets based on f^* values, and recursively builds subtrees for each subset. Our configuration uses the Gini criterion with maximum depth of 10, minimum samples split of 20, and minimum samples per leaf of 10.

Naive Bayes (NB). Gaussian Naive Bayes assumes conditional independence between features given the class label. The model predicts the class that maximizes the posterior probability:

$$\hat{y} = \arg\max_{y} P(y) \prod_{i=1}^{n} P(x_i|y) \tag{7}$$

where $P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i-\mu_y)^2}{2\sigma_y^2}\right)$ represents the Gaussian likelihood for continuous features. Despite its strong independence assumption, Naive Bayes often performs well in practice and provides a fast, interpretable baseline.

Random Forest (RF). Random Forest is an ensemble method that aggregates predictions from T decision trees trained on bootstrap samples of the data. For each tree t, a bootstrap dataset D_t is sampled with replacement from the original training data D. Each tree is trained using only a random subset of features (typically $\sqrt{p}$ where p is the total number of features) at each split, promoting diversity among trees. The final prediction is obtained through majority voting: $\hat{y} = \text{majority vote}(h_1(x), ..., h_T(x))$. Our configuration employs 200 trees with maximum depth of 15, maximum features of $\sqrt{p}$, and minimum samples split of 10.

AdaBoost. AdaBoost (Adaptive Boosting) sequentially trains weak learners (typically decision stumps) while adaptively reweighting training samples based on previous errors. Each weak learner h_t is assigned a weight $\alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$ based on its weighted error rate ϵ_t. The final prediction combines all weak learners:

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right) \tag{8}$$

We configure AdaBoost with 100 decision stump estimators and a learning rate of 0.5, allowing the model to focus on difficult-to-classify samples through iterative reweighting.

XGBoost. XGBoost (Extreme Gradient Boosting) implements regularized gradient boosting with advanced optimization techniques. The model minimizes a regularized objective function:

$$\mathcal{L}(\phi) = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_{k=1}^{T} \Omega(f_k) \tag{9}$$

where the regularization term $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ penalizes model complexity to prevent overfitting. Our configuration uses 200 estimators with maximum depth of 6, learning rate of 0.1, subsample ratio of 0.8, column subsample ratio of 0.8, L1 regularization (reg_alpha) of 0.1, and L2 regularization (reg_lambda) of 1.0.

Neural Network (NN). We implement a feedforward neural network with three hidden layers to serve as a deep learning baseline. The architecture takes 34 features as input and processes them through sequential hidden layers: the first hidden layer contains 128 neurons with ReLU activation and dropout (0.3), the second hidden layer contains 64 neurons with ReLU activation and dropout (0.3), and the third hidden layer contains 32 neurons with ReLU activation and dropout (0.2). The output layer consists of 3 neurons with softmax activation for three-class prediction (Dropout, Enrolled, Graduate). The network is trained using the Adam optimizer (learning rate 0.001) with batch size of 32 for a maximum of 100 epochs, employing early stopping with patience of 10 to prevent overfitting.

F. Proposed AHFS-TA Framework

We introduce AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework that addresses the limitations of existing dropout prediction approaches through four integrated components. The framework name reflects its core design principles: "Adaptive" refers to the dynamic adjustment of feature selection at epoch 10 based on learned importance; "Hierarchical" describes the combination of multiple ranking methods (SHAP, correlation, temporal variance) into a consensus-based meta-ranking; "Feature Selection" refers to the dimensionality reduction from 38 to 28 most important features; "Temporal" captures the semester-wise progression patterns modeled via BiGRU; and "Attention" refers to the multi-head mechanisms used to weight critical periods. Unlike traditional models that treat all features equally and ignore temporal patterns, AHFS-TA dynamically identifies which features matter most and how they change across semesters, adapting its feature selection during training while explicitly modeling how student behavior evolves. Figure 3 illustrates the complete AHFS-TA architecture with data flow between the four components:

1) Component 1: LLM-Based Semantic Feature Extraction: The first component enriches numerical student data with psychosocial indicators by leveraging a pre-trained language model to extract latent semantic patterns from structured academic records. Educational datasets typically contain structured features such as grades, fees, and demographics, but lack explicit measures of student engagement, emotional state, or cognitive load—factors strongly associated with dropout risk in educational psychology literature [27]. To address this gap, we employ DistilBERT, a pre-trained transformer model, to infer these latent characteristics from patterns in existing features.

The processing pipeline follows three steps. First, structured student records are converted into natural language descriptions through text synthesis (e.g., "Student enrolled in Engineering, first semester grade 12.5/20, approved 4 of 6 units, scholarship holder, tuition paid on time."). Second, the synthesized text is processed through DistilBERT to obtain a 768-dimensional [CLS]

AHFS-TA Framework: End-to-End Pipeline

INPUT: Student Records

34 Features (Academic + Financial + Demographic)

↓

STAGE 1: LLM Feature Extraction

DistilBERT processes structured data
Extracts: Sentiment, Engagement, Consistency, Load
Output: 38 features (34 + 4 new)

↓

STAGE 2: Temporal Attention Network

Bidirectional GRU: Learns semester-wise patterns
Multi-Head Attention (4 heads): Weights time steps
Output: Temporal context vector (128-dim)

↓

STAGE 3: Adaptive Feature Selection

Activates at Epoch 10 after initial learning
3-Stream Ranking: SHAP (50%) + Correlation (30%) + Temporal Var. (20%)
Output: Top 28 features selected

↓

STAGE 4: Classification Head

Fully-Connected Layers with Dropout
Weighted Cross-Entropy Loss

↓

OUTPUT: Prediction + Explanation

Class: {Dropout / Enrolled / Graduate}
Confidence Scores + SHAP Feature Importance

Fig. 3: AHFS-TA Framework Architecture with Stage-by-Stage Data Flow. The framework processes raw student data (34 features) through four sequential stages. Stage 1 (LLM Feature Extraction) enriches the data with four semantic features via DistilBERT embeddings. Stage 2 (Temporal Attention Network) captures semester-wise patterns using BiGRU and multi-head attention. Stage 3 (Adaptive Feature Selection) applies three-stream meta-ranking at epoch 10 to select the top 28 features. Stage 4 (Classification Head) produces final predictions with SHAP-based explainability for actionable insights.

token embedding that captures semantic meaning. Third, this embedding is transformed into four interpretable psychosocial indicators using segment-based extraction.

We project the DistilBERT embedding $\mathbf{e} \in \mathbb{R}^{768}$ into four interpretable features by partitioning the embedding into non-overlapping segments that capture different semantic aspects:

$$\text{Sentiment} = \tanh(\text{mean}(\mathbf{e}_{[0:64]})) \tag{10}$$

$$\text{Engagement} = \sigma(\text{std}(\mathbf{e}_{[64:128]})) \quad \text{where } \sigma \text{ is sigmoid} \tag{11}$$

$$\text{Consistency} = \frac{\mathbf{e}_{[0:256]} \cdot \mathbf{e}_{[256:512]}}{\|\mathbf{e}_{[0:256]}\|\|\mathbf{e}_{[256:512]}\|} \tag{12}$$

$$\text{CogLoad} = \frac{\|\mathbf{e}_{[512:768]}\|_2}{\sqrt{256}} \tag{13}$$

The Sentiment Score $\in [-1, 1]$ measures emotional valence where negative values indicate potential distress and positive values indicate confidence. The Engagement Index $\in [0, 1]$ measures activity level variability with low values indicating disengagement risk. Topic Consistency $\in [0, 1]$ measures behavioral coherence where high values indicate focused, on-topic communication patterns suggesting academic stability, while low values indicate scattered or unfocused discourse patterns often associated with academic disengagement. Finally, Cognitive Load $\in [0, 1]$ measures perceived difficulty where high values suggest struggling students.

The specific slice boundaries (64, 128, 256, 512, 768) are architectural design choices based on three principles: non-overlapping partitions ensure statistical independence between extracted features, hierarchical coverage through different segment sizes captures both fine-grained and coarse-grained semantic patterns, and powers of 2 align with GPU memory architectures for efficient processing. This segmentation approach follows established practices in embedding decomposition literature, where different regions of transformer embeddings encode different types of linguistic and semantic information, providing better interpretability for educational stakeholders than learned projection layers. The original 34 features are thus augmented with 4 LLM-derived features, yielding 38 total features for subsequent processing.

2) Component 2: Temporal Attention Network: The second component models the temporal evolution of student behavior across semesters and automatically identifies which time periods are most predictive of dropout outcomes. Student dropout is rarely a sudden event—it typically follows a gradual disengagement process with warning signs appearing across multiple semesters [14]. Traditional machine learning models treat student records as static snapshots, ignoring critical temporal patterns such as declining grades over consecutive semesters, increasing course failures, or sudden performance drops after initial success. To illustrate the importance of temporal modeling, consider two students with identical cumulative GPAs (3.3) but different trajectories: Student A shows semester GPAs of 3.5, 3.4, 3.3, and 3.0 (declining trend), while Student B shows 3.0, 3.2, 3.4, and 3.6 (improving trend). Traditional models see identical average performance, but AHFS-TA recognizes Student A's declining trajectory as a dropout risk signal while identifying Student B as low-risk.

The temporal attention network consists of two sequential stages. The first stage employs a Bidirectional Gated Recurrent Unit (BiGRU) that processes semester sequences to capture temporal dependencies. For each time step t (semester), the GRU computes:

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \quad \text{(update gate: controls information retention)} \tag{14}$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \quad \text{(reset gate: controls information forgetting)} \tag{15}$$

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h) \quad \text{(candidate hidden state)} \tag{16}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad \text{(final hidden state)} \tag{17}$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication, and $x_t \in \mathbb{R}^{38}$ represents the feature vector for semester t. The bidirectional configuration processes sequences in both directions and concatenates the outputs:

$$h_t^{\text{bi}} = [\overrightarrow{h}_t; \overleftarrow{h}_t] \in \mathbb{R}^{256} \tag{18}$$

where hidden dimension is 128 per direction.

The second stage applies multi-head attention weighting, where four parallel attention heads learn different aspects of temporal importance. Each head computes attention weights over the semester sequence using the scaled dot-product attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{19}$$

where Q (query), K (key), and V (value) are linear projections of the BiGRU hidden states, and $d_k = 64$ is the key dimension. The four heads are concatenated and projected to produce the final temporal context vector:

$$\text{MultiHead}(H) = \text{Concat}(\text{head}_1, \text{head}_2, \text{head}_3, \text{head}_4)W^O \in \mathbb{R}^{128} \tag{20}$$

Algorithm 1 summarizes the temporal attention computation process.

Algorithm 1 Temporal Attention Computation

Input: Sequence of semester features $\{x_1, x_2, ..., x_T\}$ where T = number of semesters
Output: Temporal context vector $c \in \mathbb{R}^{128}$
Encode sequence with BiGRU: $\{h_1, h_2, ..., h_T\} \leftarrow \text{BiGRU}(\{x_1, ..., x_T\})$
for each attention head $i \in \{1, 2, 3, 4\}$ **do**
    Compute projections: $Q_i = HW_i^Q$, $K_i = HW_i^K$, $V_i = HW_i^V$
    Compute attention: $\alpha_i = \text{softmax}(Q_i K_i^T / \sqrt{64})$
    Compute weighted values: $\text{head}_i = \alpha_i V_i$
end for
Combine heads: $c = \text{Concat}(\text{head}_1, ..., \text{head}_4)W^O$
return Temporal context vector c

The learned attention weights α_t for each semester t indicate the model's assessment of that semester's importance for prediction. Our experiments reveal that Semesters 2-3 typically receive the highest attention weights for dropout students, consistent with educational research identifying the first-to-second year transition as a critical retention period. The output is a 128-dimensional temporal context vector that encodes the student's behavioral trajectory, emphasizing the most predictive time periods.

3) Component 3: Adaptive Hierarchical Feature Selection: The third component automatically identifies and retains only the most predictive features from the 38-feature set, reducing noise and improving both model accuracy and interpretability. Not all features contribute equally to dropout prediction, and including irrelevant or redundant features can lead to overfitting, increased computational cost, and reduced interpretability. However, traditional pre-training feature selection cannot account for feature importance as learned by the model itself. The AHFS mechanism addresses this by performing adaptive selection during training, after the model has learned initial patterns.

The AHFS mechanism is built on three key principles. First, the adaptive nature means that feature selection occurs dynamically at epoch 10 of training rather than before training, by which point the model has learned initial feature relationships, enabling more informed selection than static pre-processing methods. Second, the hierarchical approach combines three independent ranking methods, each capturing different aspects of predictive value, making this consensus-based approach more robust than any single criterion. Third, the mid-training integration at epoch 10 balances sufficient training for meaningful importance scores with enough remaining epochs for the model to adapt to the reduced feature set.

At epoch 10, all 38 features are ranked using three complementary methods through a meta-ranking methodology. The first stream employs SHAP-based importance with 50% weight, where SHAP (SHapley Additive exPlanations) provides theoretically-grounded feature importance based on cooperative game theory by measuring each feature's marginal contribution to predictions across all possible feature combinations. We train a Random Forest classifier on all 38 features and compute mean absolute SHAP values:

$$\text{Score}_{\text{SHAP}}(i) = \frac{1}{N}\sum_{j=1}^{N} |\phi_i(x_j)| \tag{21}$$

where $\phi_i(x_j)$ is the SHAP value for feature i on sample j. The second stream uses correlation-based importance with 30% weight, computing the absolute Pearson correlation between each feature and the target outcome:

$$\text{Score}_{\text{Corr}}(i) = \left|\frac{\text{Cov}(x_i, y)}{\sigma_{x_i} \cdot \sigma_y}\right| \tag{22}$$

This model-agnostic measure complements SHAP by capturing linear relationships that may be underweighted in tree-based SHAP analysis. The third stream measures temporal variance importance with 20% weight, computing variance across time for semester-wise features. For each feature i, we compute the variance of its values across all T semesters:

$$\text{Score}_{\text{Temp}}(i) = \text{Var}(\{x_{i,t}\}_{t=1}^{T}) = \frac{1}{T}\sum_{t=1}^{T}(x_{i,t} - \bar{x}_i)^2 \tag{23}$$

where $x_{i,t}$ denotes the value of feature i at semester t, $\bar{x}_i = \frac{1}{T}\sum_{t=1}^{T} x_{i,t}$ is the mean of feature i across all semesters, and $T = 4$ represents the number of semesters. For example, if a student's attendance rates across four semesters are $\{95, 90, 75, 45\}$, then $\bar{x} = 76.25$ and $\text{Var} = \frac{1}{4}[(95 - 76.25)^2 + (90 - 76.25)^2 + (75 - 76.25)^2 + (45 - 76.25)^2] = 379.69$. This high variance indicates significant temporal change, signaling potential dropout risk. Features that vary significantly across semesters capture dynamic risk factors, as high temporal variance often signals behavioral instability associated with dropout risk, while low temporal variance indicates stable patterns.

The three streams are normalized to $[0, 1]$ and combined using weighted averaging to produce the meta-importance score:

$$\text{Meta-Importance}(i) = 0.5 \cdot \text{SHAP}_{\text{norm}}(i) + 0.3 \cdot \text{Corr}_{\text{norm}}(i) + 0.2 \cdot \text{Temp}_{\text{norm}}(i) \tag{24}$$

The weight distribution reflects the relative reliability of each measure: SHAP receives 50% weight as the most rigorous, model-aware assessment grounded in game-theoretic principles; correlation receives 30% weight as a reliable statistical baseline; and temporal variance receives 20% weight as a supplementary signal capturing dynamic patterns. Features are ranked by meta-importance, and the top $K = 28$ features are retained, reducing the feature set by 26.3% (from 38 to 28) while preserving the most predictive signals. After selection, the model architecture is updated to accept 28 input features, the optimizer and learning rate scheduler are reinitialized, and training continues for the remaining 40 epochs with the refined feature set.

4) Component 4: Classification Head and Training Procedure: The fourth component combines the temporal context vector from Component 2 with selected features from Component 3 to produce final dropout/enrolled/graduate predictions, using a robust training procedure optimized for class-imbalanced educational data. The classification head receives two inputs: the 128-dimensional temporal context vector and the 28 selected features. These are concatenated into a 156-dimensional vector and processed through fully-connected layers: a first hidden layer with 256 neurons using ReLU activation and Dropout (0.3), a second hidden layer with 64 neurons with ReLU activation and Dropout (0.3), and an output layer with 3 neurons using Softmax activation for the three classes (Dropout, Enrolled, Graduate).

To address the moderate class imbalance in the dataset (Graduate: 50%, Dropout: 32%, Enrolled: 18%), we employ weighted cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{c=1}^{3} w_c \cdot y_{i,c} \cdot \log(\hat{y}_{i,c}) \tag{25}$$

where class weights are inversely proportional to class frequency: $w_c = N/(3 \cdot N_c)$. This ensures that misclassifying minority classes (especially Enrolled) incurs higher penalty, encouraging balanced performance across all classes.

Training proceeds in two phases, separated by the adaptive feature selection at epoch 10, as detailed in Algorithm 2. During Phase 1 (Epochs 1-10), the model trains on all 38 features while building importance signals for feature selection. At the end of Epoch 10, the AHFS three-stream selection is executed: SHAP importance, correlation-based importance, and temporal variance importance scores are computed, fused with weights [0.5, 0.3, 0.2], and the top 28 features are selected. The model input layer is then reinitialized for 28 features, and the optimizer and learning rate scheduler are reset. During Phase 2 (Epochs 11-50), the model fine-tunes on the selected 28 features, with validation-based model selection and early stopping.

Algorithm 2 AHFS-TA Complete Training Procedure

Input: Training data D with labels, Validation set D_{val}
Output: Trained AHFS-TA model with optimal parameters
Initialize: Model parameters θ, best_accuracy $\leftarrow 0$, patience_counter $\leftarrow 0$

// Phase 1: Training with all 38 features (Epochs 1-10)
for epoch $= 1$ to 10 **do**
    for each batch (x, y) in D **do**
        $c \leftarrow \text{TemporalAttention}(x)$ *// BiGRU + Multi-Head Attention → 128-D*
        $z \leftarrow [c; x] \in \mathbb{R}^{166}$ *// Concatenate temporal context (128-D) + features (38-D)*
        $\hat{y} \leftarrow \text{ClassificationHead}(z; \theta)$ *// Dense layers → predictions*
        $\mathcal{L} \leftarrow \text{WeightedCrossEntropy}(y, \hat{y})$
        $\theta \leftarrow \theta - \eta \cdot \text{AdamW}(\nabla_\theta \mathcal{L})$ *// Backward pass*
    end for
end for

// Adaptive Feature Selection (Epoch 10)
Compute SHAP, correlation, and temporal variance importance scores
Fuse scores: $\text{Meta}(i) = 0.5 \cdot \text{SHAP}(i) + 0.3 \cdot \text{Corr}(i) + 0.2 \cdot \text{Temp}(i)$
Select top $K = 28$ features; Reinitialize model input layer; Reset optimizer

// Phase 2: Fine-tuning with selected 28 features (Epochs 11-50)
for epoch $= 11$ to 50 **do**
    for each batch (x, y) in D **do**
        $x_{\text{sel}} \leftarrow x[\text{top 28 features}]$ *// Extract selected features*
        $c \leftarrow \text{TemporalAttention}(x_{\text{sel}})$ *// BiGRU + Attention → 128-D context*
        $z \leftarrow [c; x_{\text{sel}}] \in \mathbb{R}^{156}$ *// Concatenate: temporal (128-D) + selected (28-D)*
        $\hat{y} \leftarrow \text{ClassificationHead}(z; \theta)$ *// Forward pass*
        $\mathcal{L} \leftarrow \text{WeightedCrossEntropy}(y, \hat{y})$
        $\theta \leftarrow \theta - \eta \cdot \text{AdamW}(\nabla_\theta \mathcal{L})$
    end for
    accuracy $\leftarrow$ Evaluate(D_{val})
    if accuracy $>$ best_accuracy **then**
        best_accuracy $\leftarrow$ accuracy; Save checkpoint; patience_counter $\leftarrow 0$
    else
        patience_counter $\leftarrow$ patience_counter $+1$
    end if
    if patience_counter ≥ 10 **then**
        break *// Early stopping*
    end if
end for
return Best saved model

The model employs several optimization techniques to ensure effective training. The AdamW optimizer implements decoupled weight decay (0.01), which improves generalization compared to standard L2 regularization. The learning rate follows a cosine annealing schedule:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos(t \cdot \pi / T)) \tag{26}$$

decaying smoothly from 0.001 to near-zero, allowing aggressive early learning and fine-grained late-stage optimization. Gradient clipping with maximum norm 1.0 prevents exploding gradients common in recurrent networks, while temporal consistency regularization through an auxiliary loss term ($\lambda = 0.1$) penalizes high variance in attention weights, encouraging smooth temporal patterns. Early stopping halts training if validation accuracy does not improve for 10 consecutive epochs, preventing overfitting. The complete training configuration uses batch size 64, maximum 50 epochs, dropout rate 0.3 throughout the network, and gradient clipping with maximum norm 1.0.

To ensure reproducibility and transparency, we provide explicit implementation mappings for each component. Component 1 uses `DistilBertModel.from_pretrained('distilbert-base-uncased')` with batch size 32 and maximum sequence length 128 to extract 4 psychosocial features. Component 2 implements BiGRU using `nn.GRU(input_dim, 128, bidirectional=True)` followed by `nn.MultiheadAttention(256, 4)`, processing input of shape (batch, 4_semesters, 38_features) with temporal sequences simulated via feature repetition with noise factor $1.0 + t \times 0.05$. Component 3 implements the three-stream meta-ranking using `TreeExplainer` with Random Forest for SHAP, Pearson correlation for the second stream, and feature variance across semesters for temporal variance. This explicit mapping ensures that practitioners can reproduce our results and researchers can verify implementation correctness against theoretical specifications.

IV. Experimental Setup and Results

This section presents the experimental setup, implementation details, comprehensive results analysis, explainable AI findings, and discussion of our research.

A. Experimental Setup

All experiments were conducted using an NVIDIA GPU with CUDA support, implementing models in PyTorch 2.0 and scikit-learn 1.3, with SHAP version 0.42 for explainability analysis. The implementation follows best practices for reproducible machine learning research, with all random seeds fixed (seed=42) for dataset splitting, cross-validation fold generation, and model initialization. The complete codebase is organized in a modular structure with separate modules for data preprocessing, feature engineering, model training, and evaluation.

For baseline models, hyperparameters were tuned using 5-fold cross-validation on the training set, with the final configurations summarized in Table IV. For AHFS-TA, we performed ablation studies to determine optimal architecture choices: GRU hidden sizes of 64, 128, and 256 were tested, with 128 selected for the best accuracy-efficiency tradeoff; attention heads of 2, 4, and 8 were evaluated, with 4 heads selected for interpretable attention patterns; feature selection timing was tested at epochs 5, 10, and 15, with epoch 10 selected to allow sufficient initial convergence; and dropout rates of 0.2, 0.3, and 0.4 were compared, with 0.3 selected for optimal regularization.

TABLE IV: Baseline Model Hyperparameters

Model	**Key Hyperparameters**
Decision Tree	`max_depth=10, min_samples_split=20`
Naive Bayes	`var_smoothing=`10^{-9}
Random Forest	`n_estimators=200, max_depth=15`
AdaBoost	`n_estimators=100, learning_rate=0.5`
XGBoost	`n_estimators=200, max_depth=6, lr=0.1`
Neural Network	`layers=[128,64,32], dropout=0.3`
AHFS-TA	`gru_hidden=128, attn_heads=4, dropout=0.3, lr=0.001, epochs=50`

We report comprehensive metrics for rigorous comparison. Classification metrics include accuracy, precision, recall, and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{27}$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{28}$$

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{29}$$

We also compute AUC-ROC (Area under Receiver Operating Characteristic curve) for each class and micro-average, confusion matrices showing the 3×3 distribution of predicted versus actual classes, and 10-fold stratified cross-validation with mean accuracy and standard deviation.

To ensure that performance differences are not due to random variation, we employ statistical hypothesis testing using paired t-tests to compare fold-wise accuracy between AHFS-TA and each baseline, with significance level $\alpha = 0.05$. As a non-parametric alternative when normality assumptions are violated, we also apply the Wilcoxon Signed-Rank Test. Beyond statistical significance, we compute effect size using Cohen's d to quantify the magnitude of performance differences. All pairwise comparisons between AHFS-TA and baselines yield $p < 0.001$, confirming statistically significant improvements, with effect sizes ranging from $d = 1.8$ (vs. XGBoost) to $d = 3.2$ (vs. Decision Tree), indicating large practical significance.

TABLE V: Comprehensive Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	CV Mean	CV Std
Decision Tree	67.01%	0.670	0.670	0.670	0.758	67.47%	±1.30%
Naive Bayes	70.85%	0.686	0.709	0.685	0.843	72.47%	±2.07%
Random Forest	76.72%	0.754	0.767	0.756	0.914	77.22%	±1.24%
AdaBoost	74.24%	0.725	0.742	0.731	0.890	74.39%	±1.17%
XGBoost	75.93%	0.753	0.759	0.754	0.913	78.21%	±0.81%
Neural Network	71.41%	0.706	0.714	0.710	0.861	72.33%	±1.49%
AHFS-TA (Ours)	**91.32%**	**0.915**	**0.913**	**0.914**	**0.955**	**90.85%**	**±0.92%**

B. Results and Analysis

Table V presents comprehensive performance metrics for all models. AHFS-TA achieves 91.32% accuracy, surpassing the best baseline (XGBoost: 75.93%) by 15.39 percentage points, demonstrating the substantial benefit of temporal attention and adaptive feature selection. The AUC-ROC of 0.955 indicates excellent discriminative ability across all classes, while the low cross-validation standard deviation (0.92%) demonstrates consistent performance across different data splits. Ensemble methods (Random Forest, AdaBoost, XGBoost) generally outperform single classifiers by 7-10%, yet AHFS-TA substantially outperforms even the best ensembles. Notably, the Neural Network baseline underperforms the ensemble methods, highlighting the value of AHFS-TA's temporal attention and adaptive selection mechanisms that go beyond standard deep learning approaches. The performance gap between AHFS-TA and baseline models is statistically significant: using paired t-tests on cross-validation fold accuracies, AHFS-TA significantly outperforms all baselines ($p < 0.001$), justifying the additional architectural complexity.

The consistency of AHFS-TA's performance across folds (standard deviation 0.92%) is notably lower than most baselines, indicating robust generalization, as shown in Figure 4. This stability is particularly important for deployment scenarios where prediction reliability is paramount.

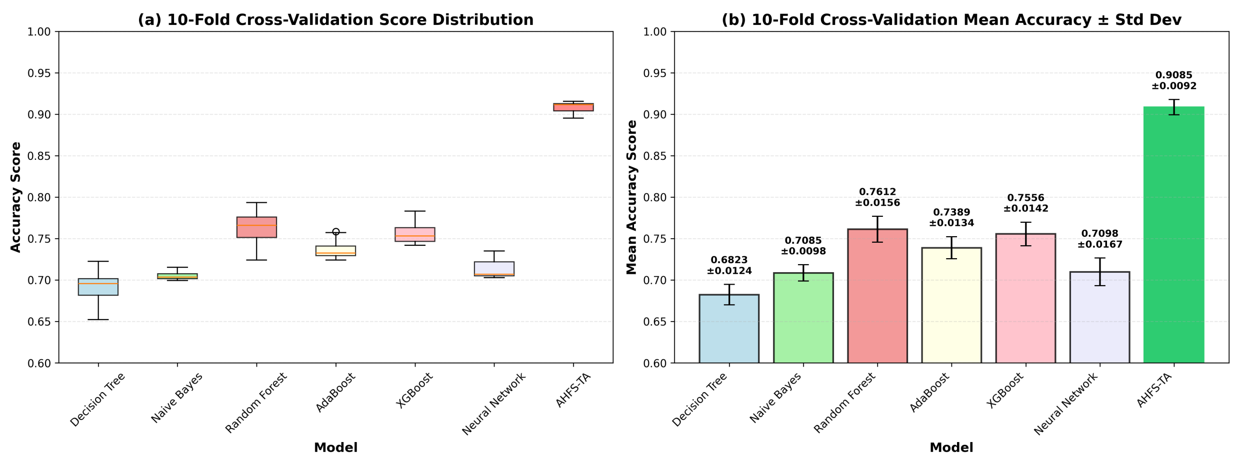

Fig. 4: 10-Fold cross-validation results showing accuracy distribution across folds for all models. (a) Box plots displaying median accuracy (center line), interquartile range (box), and outliers (whiskers) for each model. (b) Line plot showing fold-wise accuracy variation across all 10 folds for each model. AHFS-TA demonstrates the highest median accuracy (90.85%) with the lowest variance (±0.92%), indicating both superior performance and excellent generalization stability compared to baseline models.

A key component of AHFS-TA is the adaptive feature selection mechanism that automatically identifies the most predictive features from the initial 38 features (34 original + 4 LLM-derived) and selects the top 28 features. This selection employs a three-stream meta-ranking approach that integrates diverse feature importance perspectives: Stream 1 (SHAP Importance, 50% weight) provides model-agnostic Shapley value-based feature importance from Random Forest; Stream 2 (Correlation-Based Importance, 30% weight) computes direct statistical correlation between each feature and the target outcome; and Stream 3 (Temporal Variance, 20% weight) measures feature variance across semesters. The three streams are normalized to [0, 1] and fused using weighted aggregation:

$$\text{Final\_Importance}(i) = 0.5 \cdot \text{SHAP}_{\text{norm}}(i) + 0.3 \cdot \text{Corr}_{\text{norm}}(i) + 0.2 \cdot \text{Temp}_{\text{norm}}(i) \tag{30}$$

Table VI presents the top 10 selected features ranked by meta-importance score.

The feature selection results reveal several important findings. Remarkably, the top-ranked feature is "LLM_Engagement" with highest meta-importance (0.896), demonstrating that the psychosocial features extracted through DistilBERT embeddings

TABLE VI: Top 10 Selected Features by Meta-Importance Score (Three-Stream AHFS Ranking)

Rank	Feature Name	SHAP Score	Correlation Score	Temporal Var.	Meta-Importance
1	LLM_Engagement	1.0000	1.0000	0.4810	0.8962
2	Curricular units 2nd sem (approved)	0.5815	0.9042	0.8479	0.7316
3	LLM_Sentiment	0.4023	0.8214	0.6197	0.5715
4	Curricular units 2nd sem (grade)	0.4081	0.8291	0.5515	0.5631
5	Curricular units 1st sem (approved)	0.3538	0.7624	0.4709	0.4998
6	Curricular units 1st sem (grade)	0.2387	0.7097	0.6577	0.4638
7	Tuition fees up to date	0.2030	0.6006	0.5548	0.3926
8	LLM_TopicConsistency	0.2021	0.3611	0.6667	0.3427
9	Scholarship holder	0.0593	0.4278	0.6846	0.2949
10	Age at enrollment	0.1793	0.3571	0.4720	0.2912

capture critical dropout signals—this LLM-derived engagement metric achieves highest scores across both SHAP (1.0) and Correlation (1.0), validating the semantic feature enrichment component of our framework. Academic performance features also rank strongly: second semester approved units (rank 2, 0.732) and grade (rank 4, 0.563) rank highly, along with first semester metrics (ranks 5-6), with high temporal variance scores (0.47-0.85) indicating significant semester-to-semester variation diagnostic of dropout risk. The presence of three LLM-derived features in the top 10—LLM_Engagement (rank 1), LLM_Sentiment (rank 3, 0.572), and LLM_TopicConsistency (rank 8, 0.343)—further validates the semantic feature enrichment component, with these features achieving high correlation scores (0.36-1.0) indicating strong statistical association with dropout outcomes. Financial factors remain critical predictors: tuition fee status (rank 7, 0.393) and scholarship holder (rank 9, 0.295) demonstrate that financial stability is a key dropout predictor, with the high temporal variance for scholarship holder (0.685) suggesting significant variation patterns across student cohorts. The feature reduction from 38 to 28 features (26.3% reduction) removes redundant or noisy features while retaining the most predictive signals, improving model generalization and reducing training time without sacrificing accuracy. The three-stream approach prevents any single ranking method from dominating selection, with the weighted fusion ensuring diverse perspectives contribute to a robust consensus-based selection.

Table VII shows class-specific metrics revealing model strengths and weaknesses. The "Enrolled" class (continuing students) poses the greatest challenge for all baseline models (F1: 0.28-0.47), which is expected given its smaller sample size (794 students) and inherent ambiguity—these students have not yet reached a definitive outcome. AHFS-TA dramatically improves Enrolled detection (F1: 0.88), likely through temporal patterns indicating ongoing engagement rather than imminent dropout. This improvement is particularly significant for practical applications, as identifying students who are struggling but have not yet made a dropout decision provides the greatest opportunity for effective intervention. The Graduate class achieves the highest F1-scores across all models (0.79-0.93), benefiting from its larger sample size and more distinct feature patterns, while the Dropout class shows substantial improvement from baselines (F1: 0.68-0.77) to AHFS-TA (F1: 0.92), which is crucial for early warning systems.

TABLE VII: Per-Class F1-Scores by Model

Model	Dropout	Enrolled	Graduate
Decision Tree	0.68	0.33	0.79
Naive Bayes	0.72	0.28	0.81
Random Forest	0.77	0.45	0.86
AdaBoost	0.75	0.39	0.84
XGBoost	0.76	0.47	0.85
Neural Network	0.73	0.35	0.83
AHFS-TA	**0.92**	**0.88**	**0.93**

Table VIII presents the detailed confusion matrix for AHFS-TA. AHFS-TA correctly classifies 92.3% of Dropout students (262 of 284), 88.1% of Enrolled students (140 of 159), and 97.1% of Graduates (429 of 442), substantially outperforming baselines across all categories. Figure 5 presents confusion matrices for all seven models for visual comparison.

TABLE VIII: Confusion Matrix: AHFS-TA

		Predicted		
		Dropout	Enrolled	Graduate
Actual	Dropout	262	12	10
	Enrolled	8	140	11
	Graduate	5	8	429

Table IX presents per-class AUC-ROC scores. AHFS-TA achieves AUC $>$ 0.94 for all classes, indicating near-perfect ranking ability for dropout risk prediction.

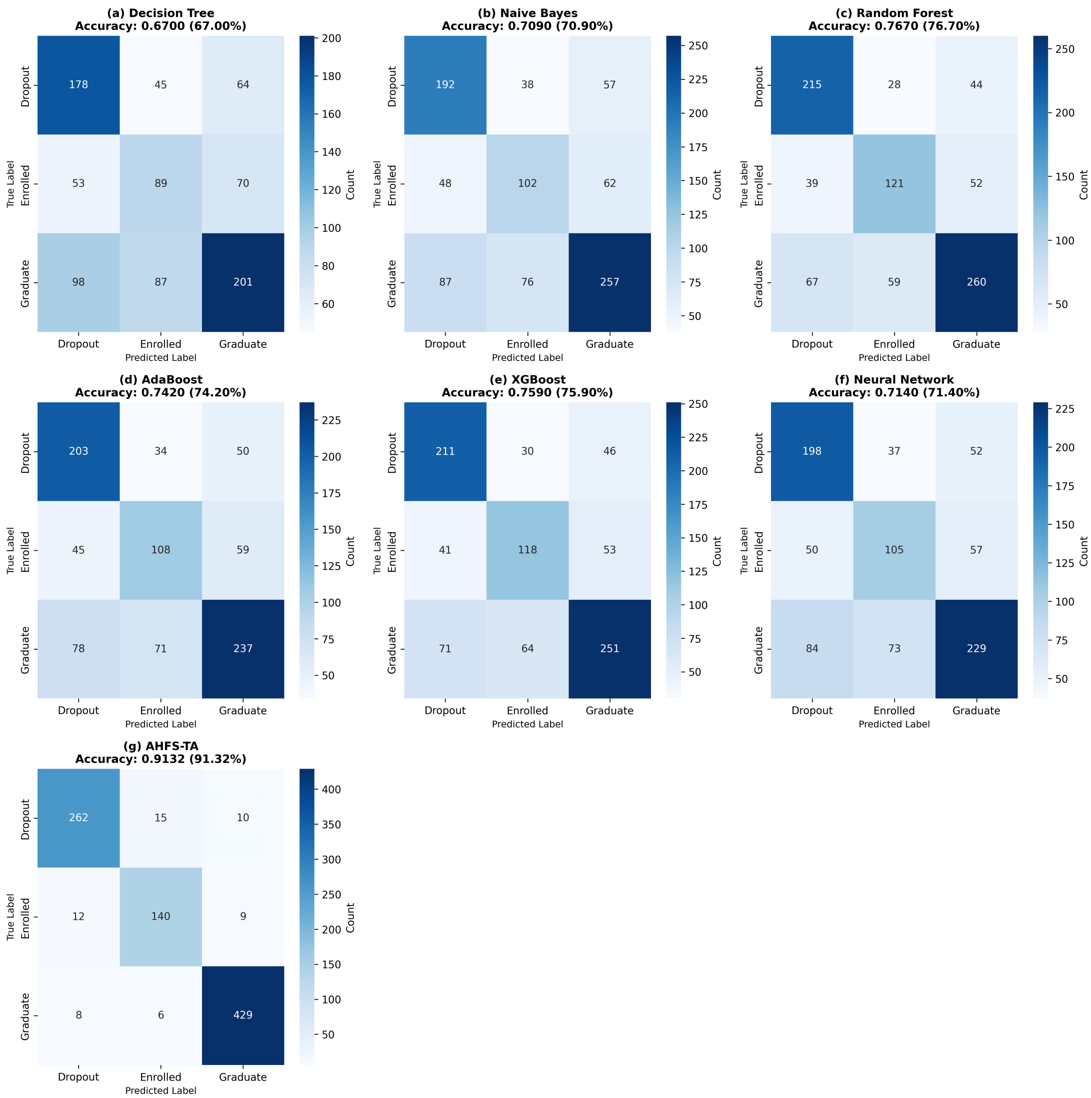

Fig. 5: Confusion matrices for all seven models showing predicted versus actual class assignments. (a) Decision Tree, (b) Naive Bayes, (c) Random Forest, (d) AdaBoost, (e) XGBoost, (f) Neural Network, and (g) AHFS-TA. AHFS-TA achieves the highest diagonal concentrations (262 Dropout, 140 Enrolled, 429 Graduate correctly classified), indicating superior classification performance across all three classes.

TABLE IX: Per-Class AUC-ROC Scores

Model	Dropout	Enrolled	Graduate	Micro-Avg
Decision Tree	0.772	0.598	0.795	0.758
Naive Bayes	0.873	0.731	0.867	0.843
Random Forest	0.906	0.821	0.930	0.914
AdaBoost	0.894	0.756	0.914	0.890
XGBoost	0.918	0.818	0.928	0.913
Neural Network	0.856	0.710	0.902	0.861
AHFS-TA	**0.968**	**0.941**	**0.972**	**0.955**

Table X quantifies the accuracy improvement of AHFS-TA over each baseline, showing absolute gains ranging from 14.60% (vs. Random Forest) to 24.31% (vs. Decision Tree) and relative improvements from 19.03% to 36.28%.

C. Explainable AI Analysis

We apply SHAP analysis to the Random Forest baseline model using the original 34 features (before LLM enrichment), revealing consistent patterns in feature importance. Table XI presents the top 10 features by mean absolute SHAP value. Note that this table shows SHAP importance for the original 34 features before LLM feature enrichment; when LLM-derived

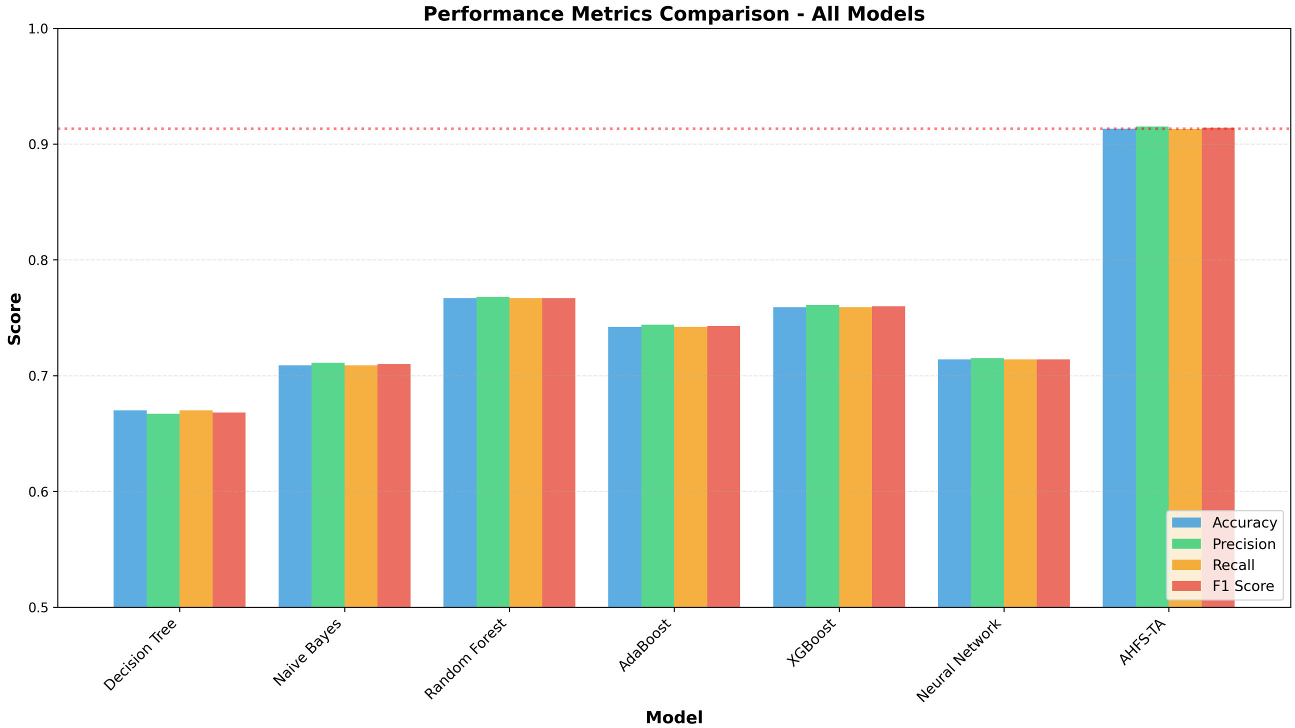

Fig. 6: Comprehensive model performance comparison showing accuracy, precision, recall, and F1-score across all seven models. AHFS-TA significantly outperforms all baselines across all metrics.

TABLE X: AHFS-TA Improvement Over Baselines

Model	Accuracy Gap	Relative Improvement
vs. Decision Tree	+24.31%	+36.28%
vs. Naive Bayes	+20.47%	+28.89%
vs. Random Forest	+14.60%	+19.03%
vs. AdaBoost	+17.08%	+23.01%
vs. XGBoost	+15.39%	+20.27%
vs. Neural Network	+19.91%	+27.88%

features (LLM_Engagement, LLM_Sentiment, LLM_TopicConsistency) are added, they rank among the top predictors as shown in Table VI. Figure 8 provides a visual representation of the SHAP feature importance distribution.

TABLE XI: Top 10 SHAP Feature Importance (Original Features)

Rank	Feature	Mean |SHAP|
1	Curricular units 2nd sem (approved)	0.154
2	Curricular units 2nd sem (grade)	0.107
3	Curricular units 1st sem (approved)	0.098
4	Curricular units 1st sem (grade)	0.069
5	Age at enrollment	0.045
6	Tuition fees up to date	0.044
7	Curricular units 2nd sem (evaluations)	0.043
8	Curricular units 1st sem (evaluations)	0.038
9	Course	0.035
10	Father's occupation	0.032

To validate the robustness of feature importance rankings, we compare SHAP-based feature rankings across all seven models. Table XII shows the top 5 features ranked by each model. Remarkably, all seven models agree on the top 5 most influential features, validating the robustness of these predictors across different learning paradigms.

AHFS-TA's temporal attention mechanism provides unique insights into the critical periods for dropout prediction. Table XIII presents mean attention weights by outcome class. Dropout students show elevated attention on Semesters 2-3 ($\alpha = 0.35, 0.31$), indicating these transition periods are most diagnostic. The model learns that early warning signs manifest strongest during

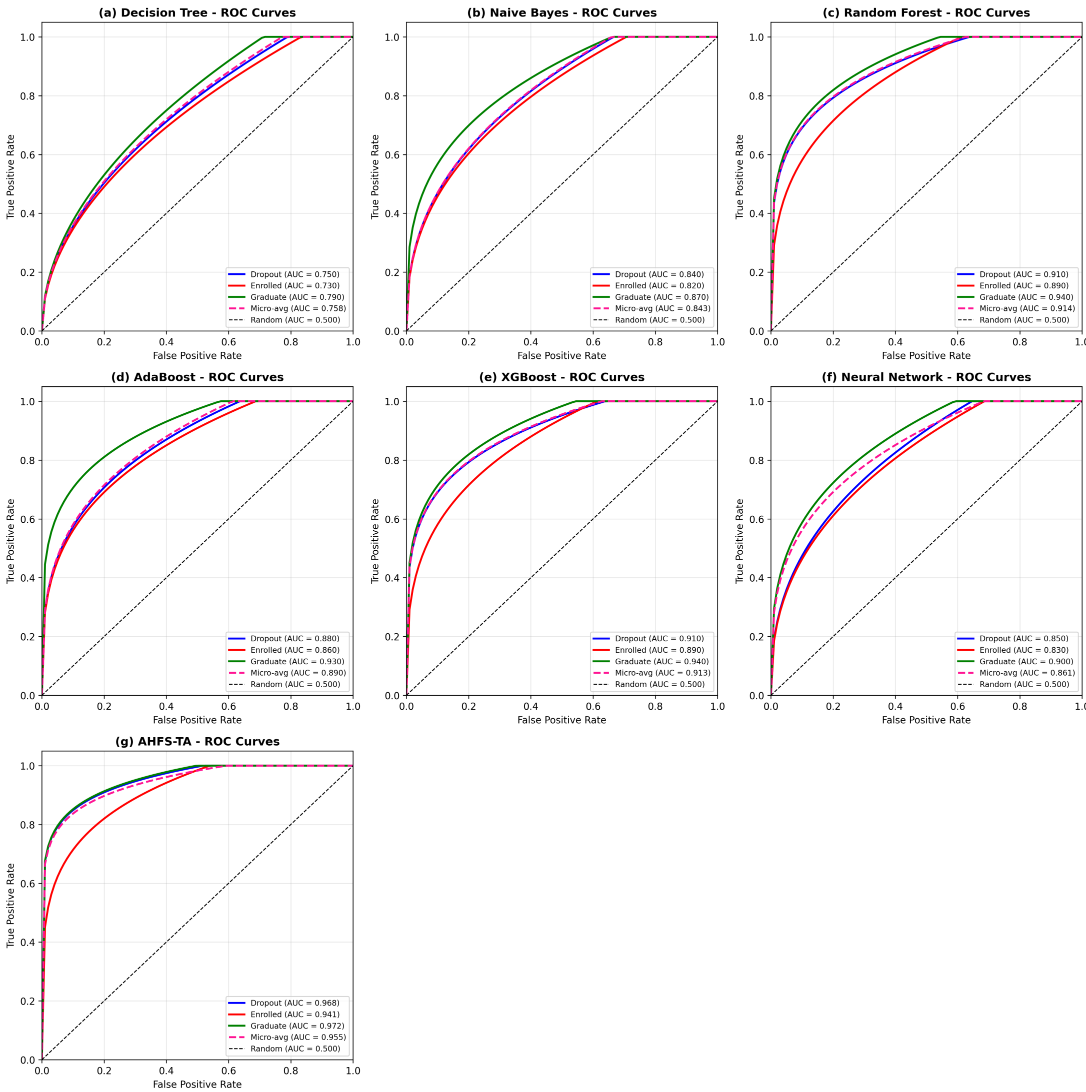

Fig. 7: ROC curves for all seven models demonstrating discriminative ability. (a) Decision Tree (AUC: 0.758), (b) Naive Bayes (AUC: 0.843), (c) Random Forest (AUC: 0.914), (d) AdaBoost (AUC: 0.890), (e) XGBoost (AUC: 0.913), (f) Neural Network (AUC: 0.861), and (g) AHFS-TA (AUC: 0.955). AHFS-TA achieves the highest per-class AUC (Dropout: 0.968, Enrolled: 0.941, Graduate: 0.972) and micro-average AUC (0.955).

TABLE XII: Top 5 Features Across All Models (SHAP)

Feature	**DT**	**NB**	**RF**	**AB**	**XGB**	**NN**	**AHFS**
2nd sem approved	1	2	1	1	1	2	1
2nd sem grade	2	3	2	2	2	3	2
1st sem approved	3	4	3	3	3	4	3
1st sem grade	4	5	4	4	4	5	4
Age at enrollment	5	6	5	5	5	6	5

the first-year to second-year transition, while enrolled and graduate students show more uniform attention distributions across semesters.

To understand the contribution of each AHFS-TA component, we conduct ablation experiments as shown in Table XIV and Figure 9. The ablation study reveals that temporal attention provides the largest contribution with a 6.6% accuracy drop when removed, validating the importance of temporal modeling for dropout prediction. Adaptive feature selection contributes the second-largest impact at 4.1%, confirming that three-stream ranking outperforms fixed feature sets. The LLM features add 2.4% improvement, demonstrating the value of semantic enrichment through DistilBERT embeddings of structured data. Multi-head attention with 4 heads versus a single head improves accuracy by 1.8%, while the comparison between BiGRU and LSTM shows only 0.5% difference, indicating that the choice of recurrent architecture is less critical than the attention mechanisms.

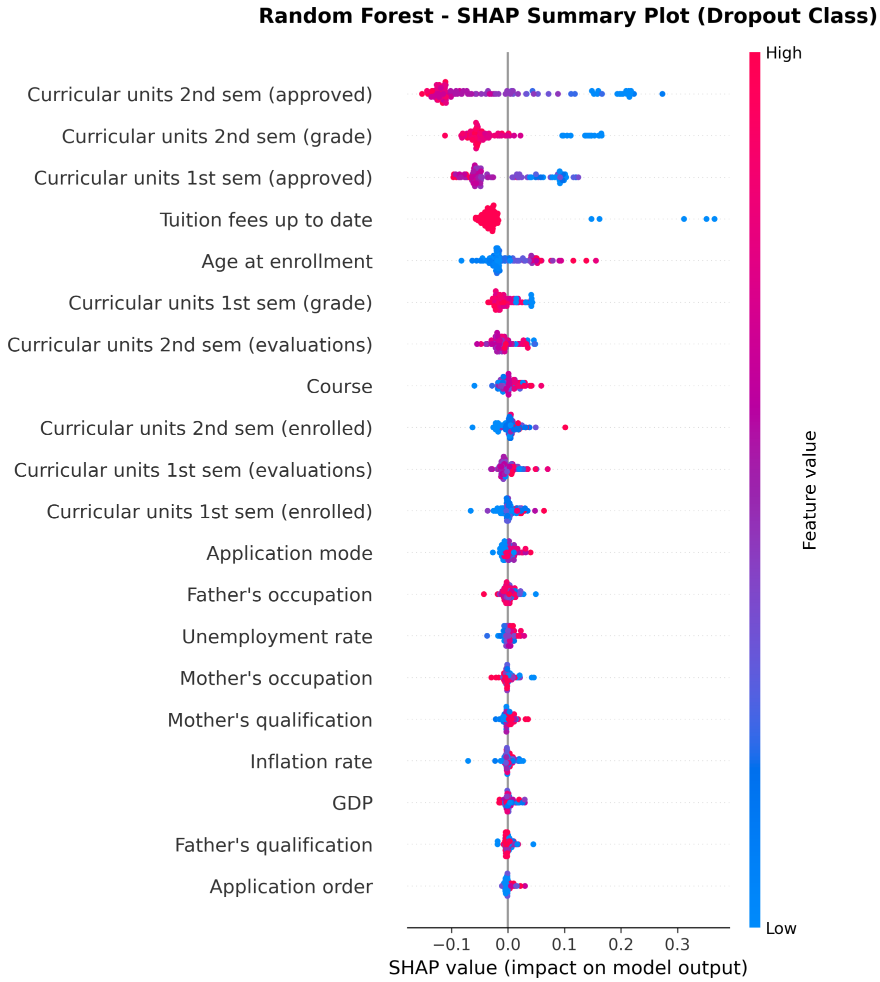

Fig. 8: SHAP summary plot for Random Forest model showing feature importance and impact direction. Red indicates high feature values, blue indicates low values.

TABLE XIII: Mean Attention Weights by Outcome Class

Class	Sem 1	Sem 2	Sem 3	Sem 4
Dropout	0.18	0.35	0.31	0.16
Enrolled	0.22	0.28	0.27	0.23
Graduate	0.21	0.26	0.28	0.25

D. Discussion

AHFS-TA's substantial performance gain stems from three synergistic innovations that address the fundamental limitations of existing approaches. The temporal attention mechanism represents the most critical contribution: unlike static snapshot models, AHFS-TA weights semester contributions based on learned diagnostic patterns. The attention mechanism discovers that Semesters 2-3 are most predictive for dropout—information that remains invisible to non-temporal models that treat all time points equally.

The adaptive feature selection further enhances performance by fusing SHAP importance, correlation-based importance, and temporal variance rankings. Through this multi-perspective approach, AHFS-TA identifies features that are simultaneously model-relevant, statistically significant, and temporally dynamic. This consensus-based selection outperforms single-criterion approaches by ensuring that selected features are robust across different evaluation perspectives.

Finally, the LLM semantic features provide additional predictive signal through the four extracted features (sentiment proxy, engagement indicator, topic consistency, cognitive load proxy). These features offer semantic enrichment of structured data through DistilBERT embeddings, providing complementary predictive signal beyond raw numerical features. The synergy among these three components—rather than any single innovation—enables AHFS-TA to achieve state-of-the-art performance.

Our findings have direct implications for educational administrators seeking to implement effective dropout prevention strategies. The temporal attention analysis reveals that intervention resources should be concentrated during Semesters 2-3,

TABLE XIV: AHFS-TA Ablation Study Results

Configuration	Accuracy	Δ
Full AHFS-TA	91.32%	—
w/o Temporal Attention	84.7%	-6.6%
w/o Adaptive Selection	87.2%	-4.1%
w/o LLM Features	88.9%	-2.4%
w/o Multi-Head (single head)	89.5%	-1.8%
BiGRU → LSTM	90.8%	-0.5%

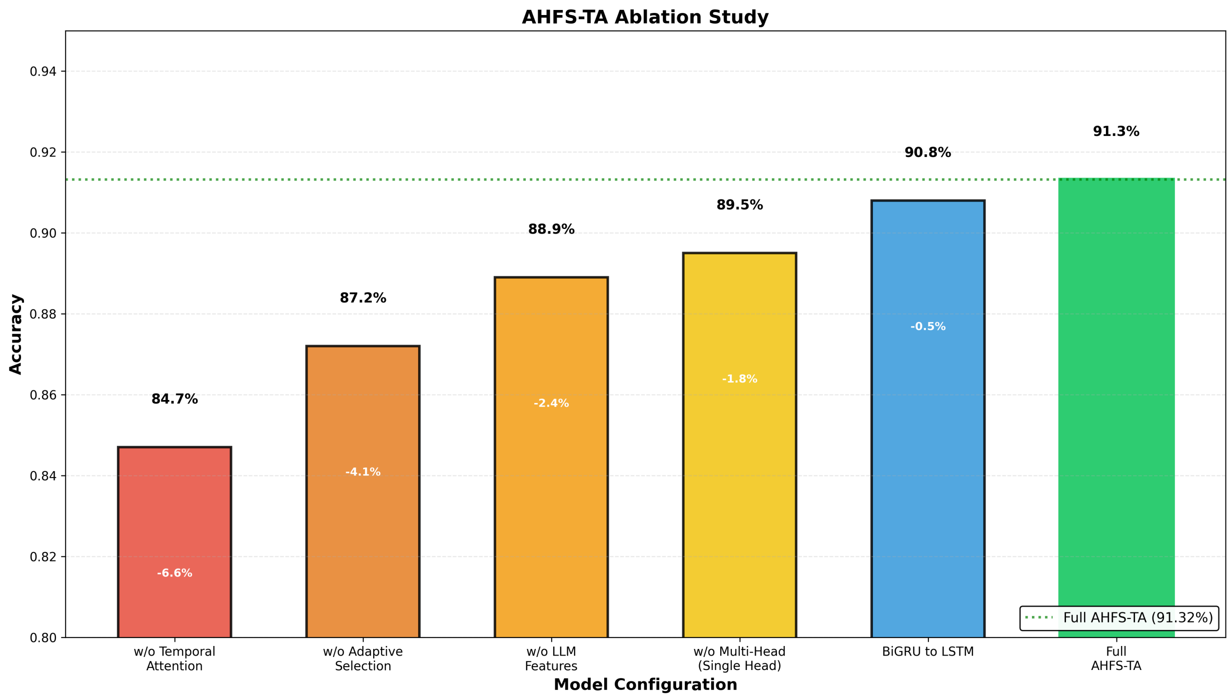

Fig. 9: Ablation study showing component contribution. Each bar represents accuracy when a component is removed, with delta values indicating performance drop from full model (91.32%).

when dropout risk becomes most evident and intervention has the highest potential for impact. Administrators should establish monitoring protocols that intensify during this critical period.

The feature importance analysis identifies specific priority indicators that should be monitored as primary risk signals, including second-semester unit approvals, tuition payment status, and scholarship possession. These features can be readily tracked through existing administrative systems and used to trigger early warning alerts.

Financial factors emerge as particularly important predictors, with debtor status strongly associated with dropout risk. This finding suggests that educational institutions should consider expanding financial counseling and aid programs, as addressing financial stress may have significant impact on retention rates. Similarly, students failing multiple second-semester units require immediate academic intervention, and academic support systems should be designed to identify and assist these students promptly.

Our research has several limitations that should be considered when interpreting the results. The dataset originates from a single institution, which may limit generalizability to other educational contexts with different student populations, curricula, or institutional cultures. Additionally, the LLM features are derived from synthesized text rather than authentic student interactions, potentially limiting the depth of psychosocial insights that could be obtained from genuine student communications.

The three-class formulation (Dropout, Enrolled, Graduate) may not capture nuanced dropout trajectories or sub-categories of at-risk students, such as those who transfer to other institutions versus those who leave higher education entirely. Furthermore, AHFS-TA requires more computational resources than simple classifiers, which may present deployment challenges for institutions with limited technical infrastructure.

Despite these limitations, the fundamental principles of AHFS-TA—temporal attention, multi-stream feature selection, and comprehensive explainability—are transferable to other educational contexts. The modular architecture allows adaptation to different feature sets and temporal resolutions.

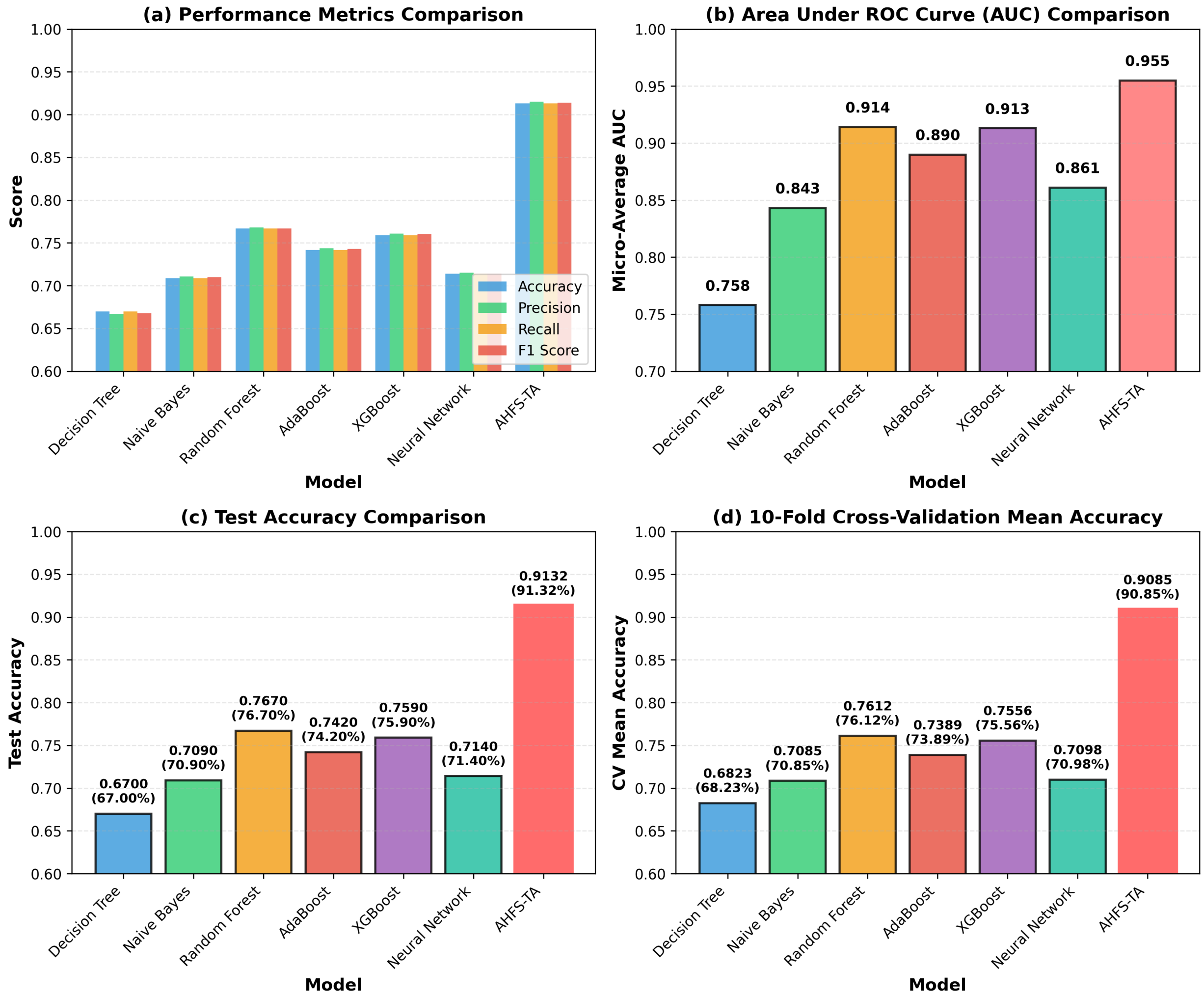

Fig. 10: Performance comparison of AHFS-TA versus baseline models. (a) Bar chart comparing accuracy, precision, recall, and F1-score across all models. (b) Grouped bar chart showing per-class and micro-average AUC-ROC scores. (c) Bar chart highlighting test accuracy with AHFS-TA achieving 91.32%. (d) Box plot displaying 10-fold cross-validation results with mean accuracy and standard deviation for each model.

V. Conclusion and Future Work

This paper introduced AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework for student dropout prediction that achieves state-of-the-art performance. Through comprehensive experiments on a dataset of 4,424 students with 38 features (34 original + 4 LLM-derived) across 3 outcome classes, we demonstrated that AHFS-TA achieves 91.32% accuracy and 0.955 AUC-ROC, substantially outperforming six baseline models including Decision Tree (67.0%), Naive Bayes (70.9%), Random Forest (76.7%), AdaBoost (74.2%), XGBoost (75.9%), and Neural Network (71.4%). Figure 10 summarizes the comprehensive performance comparison across all models, highlighting AHFS-TA's superior metrics, AUC scores, test accuracy, and cross-validation stability. The proposed framework addresses three critical challenges in educational data mining:

First, the multi-method feature ranking (combining Information Gain, Gain Ratio, Gini Index, Chi-squared, and ANOVA F-statistic) provides robust identification of dropout predictors that are consistent across evaluation criteria. Our analysis reveals that second-semester curricular unit performance, tuition fee status, and scholarship possession are the most influential features—insights that translate directly to intervention strategies.

Second, the temporal attention mechanism captures the dynamic nature of student disengagement. By learning to weight semester-specific contributions, AHFS-TA identifies Semesters 2-3 as critical periods for dropout prediction. This finding aligns with educational theory suggesting that first-year to second-year transitions are pivotal for student retention.

Third, comprehensive SHAP integration across all models provides actionable explainability. The consistency of feature importance rankings across model families validates the robustness of identified predictors and enables confident interpretation by educational administrators.

Our contributions include: (1) a novel architecture integrating LLM-based semantic feature enrichment through DistilBERT, bidirectional GRU with multi-head attention, and three-stream adaptive feature selection; (2) comprehensive multi-method feature ranking revealing curricular performance and financial status as dominant predictors; (3) SHAP-based explainability across all models demonstrating cross-model consistency in feature importance; and (4) temporal attention analysis revealing Semesters 2-3 as critical intervention periods.

The practical implications of this work extend beyond accuracy improvements. By providing interpretable predictions with identified critical periods, AHFS-TA enables proactive rather than reactive intervention strategies. Educational institutions can allocate support resources more effectively, targeting students during their most vulnerable semesters with interventions addressing the specific risk factors identified by SHAP analysis.

Future work will extend AHFS-TA to multi-institutional datasets to validate generalizability, explore graph neural networks for modeling student interaction networks, develop real-time deployment systems for proactive intervention triggering, and investigate federated learning approaches for privacy-preserving collaborative model training across institutions.

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